

Assignment -2

1. Consider a logical address space of 8 pages of 1024 words mapped into memory of 32 frames. How many bits are there in logical address and physical address?
2. Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using-First Fit, Best Fit, Worst Fit Algorithm
3. Consider the page references 7,0,1,2,0,3,0,4,2,3,0,3,2, Find the number of page fault and page hit using **OPR, FIFO, LRU and Second Chance algorithm** with 3 page frame. Analyze which is better
4. Consider following segment table:

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for following logical address?

- a. 0, 430
 - b. 1, 10
 - c. 2, 500
 - d. 3, 400
 - e. 4, 112
5. Suppose that a disk drive has 150 cylinder, numbered 0 to149. The drive is currently serving a request at cylinder 35, and previous request was at 120. The queue of pending request , in FIFO order is

8, 103, 38, 122, 10, 128, 65, and 75

Starting from the current head position, what is the total distance (in cylinder) that the disk arm moves to satisfy all pending request for each of the following disk scheduling algorithms?

- a. FCFS
- b. SSTF
- c. SCAN
- d. C-SCAN
- e. LOOK